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CNN-Based Face Mask Recognition System

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1 Problem Statement

In recent years, wearing a mask or covering the face has introduced critical vulnerabilities in standard biometric security systems. Traditional face recognition systems experience a significant drop in accuracy when facial features are obscured by masks. The aim of this project is to build a Convolutional Neural Network (CNN)-based architecture capable of simultaneously detecting mask usage and recognizing the identity of the underlying individual, ensuring reliable authentication for both masked and non-masked faces. Ultimately, the system aims to provide a robust and contact-free solution for secure access control and automated identity verification in high-stakes environments such as healthcare facilities, corporate offices, and transit hubs.

2 Objectives

1. **Design and Implementation:** To design and implement a Convolutional Neural Network (CNN)-based deep learning model to detect face mask usage and identify individuals despite facial occlusion.
2. **Feature Extraction:** To extract biometric features from the visible upper facial regions (eyes, eyebrows, and forehead) when the lower face is covered.
3. **Image and Video Processing:** To process RGB pixel matrices from static facial datasets.
4. **Mathematical Analysis:** To analyze the mathematical foundations of the chosen CNN architecture, including the optimizer, activation functions, and loss function required for multiclass verification.
5. **Performance Evaluation:** To evaluate the model's biometric performance using statistical accuracy measures such as Precision, Recall, F1-Score, and Cosine Similarity.

3 Assumptions and Constraints

1. **Biometric Saliency:** The periocular region contains sufficient distinctive information for identity verification, even when the lower half of the face is covered.
2. **Landmark Visibility:** The facial landmarks required for feature extraction are assumed to be sufficiently visible and detectable in the input images.
3. **Mesh Localization Accuracy:** The facial mesh is assumed to provide reasonably accurate landmark coordinates for extracting the relevant facial regions.
4. **Textural Occlusion Boundary:** The boundary between the mask and the visible facial region provides useful textural information for mask classification.
5. **Dataset Quality and Class Distribution:** The dataset is assumed to be cleaned and preprocessed, with corrupted, duplicate, or irrelevant images removed. Images are assumed to have relatively uniform quality, resolution, and format, with consistent labels. The *Masked* and *UnMasked* classes are assumed to be uniformly distributed.

4 Proposed Architecture

The proposed system integrates mask classification and identity recognition into a unified deep learning model. This system uses MediaPipe (developed by Google) for face detection and facion region extration, while MobileNetV2 for mask classification, and a Siamese neural network of MobileNetV2 for identity recognition.

The proposed methodology consists of two parallel classification and recognition branches to achieve the said objective mentioned in section 2. The first branch determines whether the detected face is wearing a mask, while the second branch is responsible for the identity by comparing facial features. Both branches are integrated into a unified model to produce the final output.

As illustrated in Figure 1, the proposed system integrates mask classification and identity verification into a unified model.

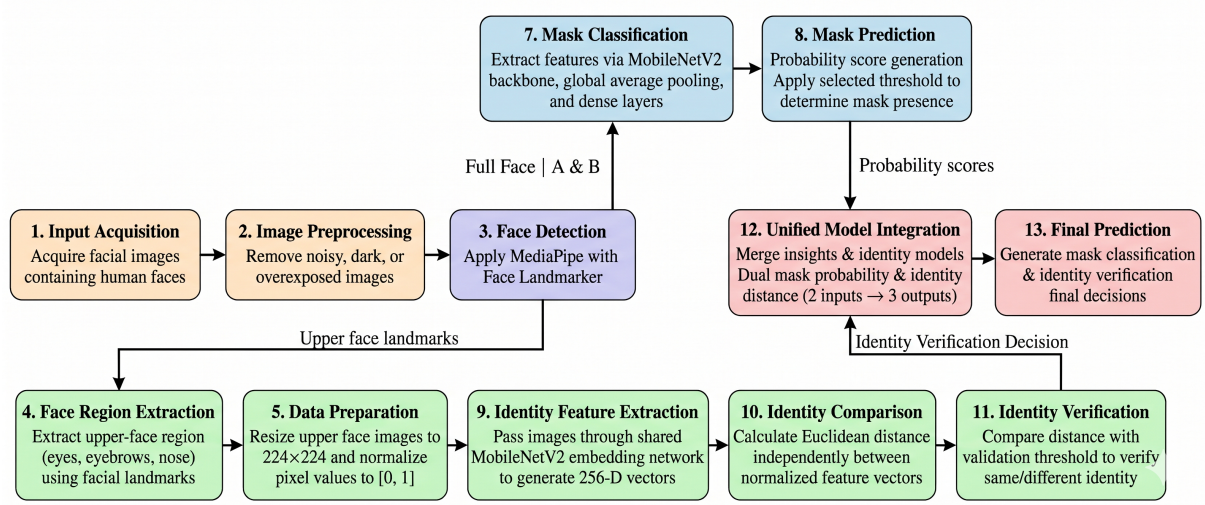


Figure 1: Proposed architecture of the CNN-based face mask recognition and identity verification system.

4.1 Step-by-Step Method

1. **Input Acquisition:** Acquire a facial image that contains human face as input to the proposed system.

2. **Image Preprocessing:** Remove any noisy, dark, or overexposed image for the dataset to ensure better prediction.

3. **Face Detection:** Apply the MediaPipe with Face Landmarker model to detect facial land markers and identify the face region.

4. **Face Region Extraction:** For identity recognition, the upper face region is extracted using relevant facial landmarks, including the eye, eyebrows and the nose region.

5. **Data Preparation:** Resize the input images to 224x224 pixels and normalize the pixel values to the range [0,1].

6. **Parallel Processing:** Pass the preprocessed facial image through two independent processing branches: the mask classification branch and the identity recognition branch.

7. **Mask Classification:** Feed the facial image into the MobileNetV2-based mask detection model. The model extracts high-level features through the MobileNetV2 backbone, followed by Global Average Pooling, Dropout, and a sigmoid-activated dense layer.

8. **Mask Prediction:** Generate a probability score using the sigmoid output. Apply the selected classification threshold to determine whether the detected person is wearing a mask or not.

9. **Identity Feature Extraction:** Pass the extracted upper-face image through the shared MobileNetV2 embedding network of the Siamese architecture to generate 256-dimentional feature representation.

10. **Identity Comparison:** Provide two upper-face images of the same person to the Siamese Network. The Network independently generates normalized feature vectors for both images. Calculate the Euclidean distance between the two normalized feature vectors to measure the similarity.

11. **Identity Verification:** Compare the cosine similarity with a threshold selected using validation data. A distance below the threshold indicates a potential identity match, whereas a distance above the threshold indicates different identities.

12. **Unified Model Integration:** Integrate the mask detection model and Identity matching model into a single unified model with two input and three output.

13. **Final Prediction:** Generate the mask classification probability for two input images and identity distance of those images, then apply decision thresholds to obtain the final predictions.

4.2 Algorithms

The proposed system consists of two major algorithmic stages:

The first stage performs image preprocessing, including removal of noisy, extreme dark or over-exposed images, image resizing and image normalization to maintain a consistency across the dataset along with that train test splitting. This stage also includes upper face region extraction, which serves as the Region of Interest(ROI) for identify verification. The detailed procedure for the first stage is mentioned below in Algorithm 1.

Algorithm 1 Image Preprocessing and Upper-Face Region Extraction

Require: Dataset containing Masked and UnMasked facial images
Ensure: Quality-filtered dataset, train/validation/test splits, normalized images, and upper-face crops

Initialize identity groups and rejected-image list
for all images in the Masked and UnMasked classes **do**
    Read image using OpenCV
    if image is unreadable **then**
        Add image to rejected-image list
        continue
    end if
    Convert image to grayscale
    Apply Gaussian blur and estimate noise from the residual
    Calculate image brightness
    if noise score > 15 or brightness < 50 or brightness > 220 **then**
        Reject image as low quality
    else
        Extract identity label from image filename
        Add image to the corresponding identity group
    end if
end for
Split identities into training (70%), validation (15%), and testing (15%)
Ensure that identity sets are mutually disjoint
Copy images into their corresponding split and mask-class directories
for all images in each dataset split **do**
    Resize image to 224×224
    Convert pixel values to floating point
    Normalize pixel values to $[0, 1]$
    Store the normalized image for mask classification
    Convert original image from BGR to RGB
    Apply MediaPipe Face Landmarker
    if facial landmarks are detected **then**
        Extract eye and nose landmark coordinates
        Determine upper-face crop boundaries with padding
        Crop the upper-face region
    else
        Use the full image as the fallback region
    end if
    Resize upper-face region to 224×224
    Save the crop in the identity-image cache
end for
return Preprocessed datasets and upper-face image cache

The second stage performs CNN based face mask classification with Siamese-network-based identity verification. The mask classification branch is used to predict whether each input is masked , while the Siamese network compares the extracted upper-face features to determine whether the two images belong to the same individual. The detailed procedure for the first stage is mentioned below in Algorithm 2.

Algorithm 2 CNN-Based Face Mask Classification and Identity Verification

Require: Two facial images A and B, trained mask classifier M, shared Siamese embedding network E, mask threshold τ_m, identity threshold τ_i

Ensure: Mask predictions for images A and B, and identity verification result

Preprocess images A and B using Algorithm 1

Mask Classification Branch

$p_A \leftarrow M(A)$

$p_B \leftarrow M(B)$

$m_A \leftarrow \begin{cases} 1, & \text{if } p_A \geq \tau_m, \\ 0, & \text{otherwise} \end{cases}$

$m_B \leftarrow \begin{cases} 1, & \text{if } p_B \geq \tau_m, \\ 0, & \text{otherwise} \end{cases}$

Identity Verification Branch

Extract upper-face regions U_A and U_B using facial landmarks

$z_A \leftarrow E(U_A)$

$z_B \leftarrow E(U_B)$

Normalize embeddings z_A and z_B using L2 normalization

$d \leftarrow \|z_A - z_B\|_2$

$s \leftarrow z_A^\top z_B$ ▷ Cosine similarity of normalized embeddings

if $s \geq \tau_i$ **then**

    $v \leftarrow$ Same Identity

else

    $v \leftarrow$ Different Identities

end if

return p_A, p_B, m_A, m_B, d, s, and v

4.3 CNN and Siamese Network Architecture

4.3.1 MobileNetV2

MobileNetV2 is used as the Convolutional Neural Network (CNN) architecture for the proposed system. It is a lightweight deep learning model designed to provide efficient image feature extraction with very low computational complexity.
Figure 2 illustrates the flowchart architecture of MobileNetV2.

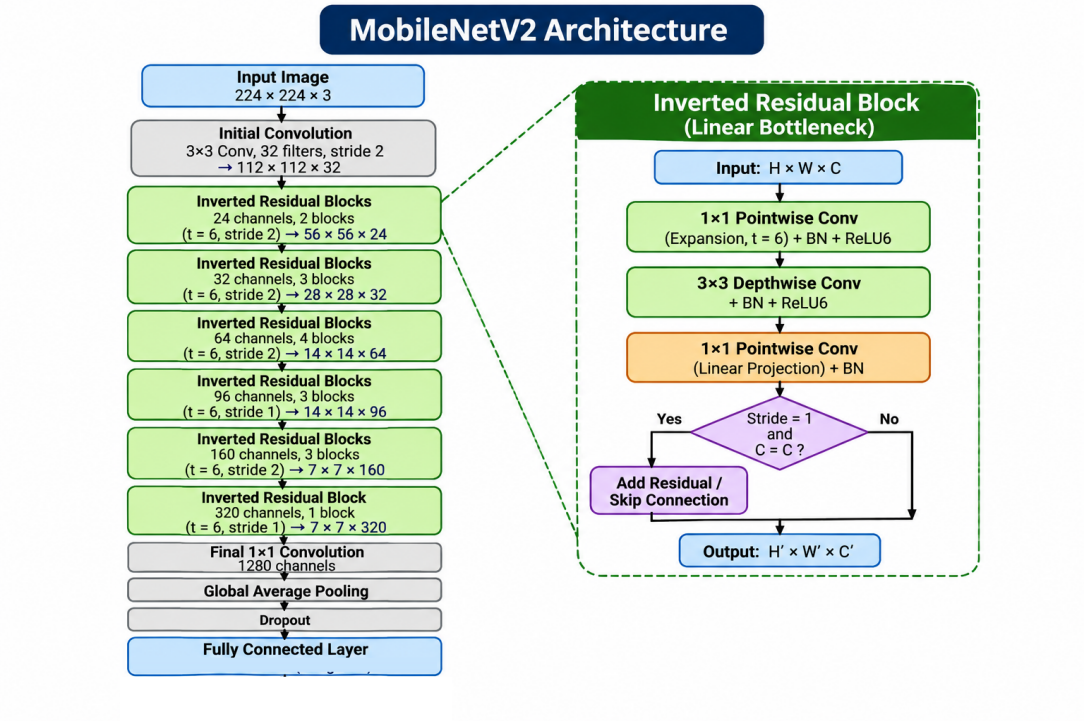


Figure 2: Flowchart architecture of MobileNetV2

4.3.2 Siamese Network with MobileNetV2

The Siamese Network consists of two identical subnetworks that share the same weights and parameters. In the proposed system, MobileNetV2 is used as the shared feature extraction backbone for both mask classification and identity verification branches.
The network receives a pair of face region images and independently extracts feature embedding using the shared MobileNetV2 model. The embedding are the compared using a distance metric to determine the similarity between the two input images. A smaller distance indicates that the images are more likely to belong to the same identity, while the larger distance indicates that they are likely to belong to different identities.
Figure 3 shows the Siamese Network architecture incorporating MobileNetV2 as the shared feature extractor and contrastive loss for training.

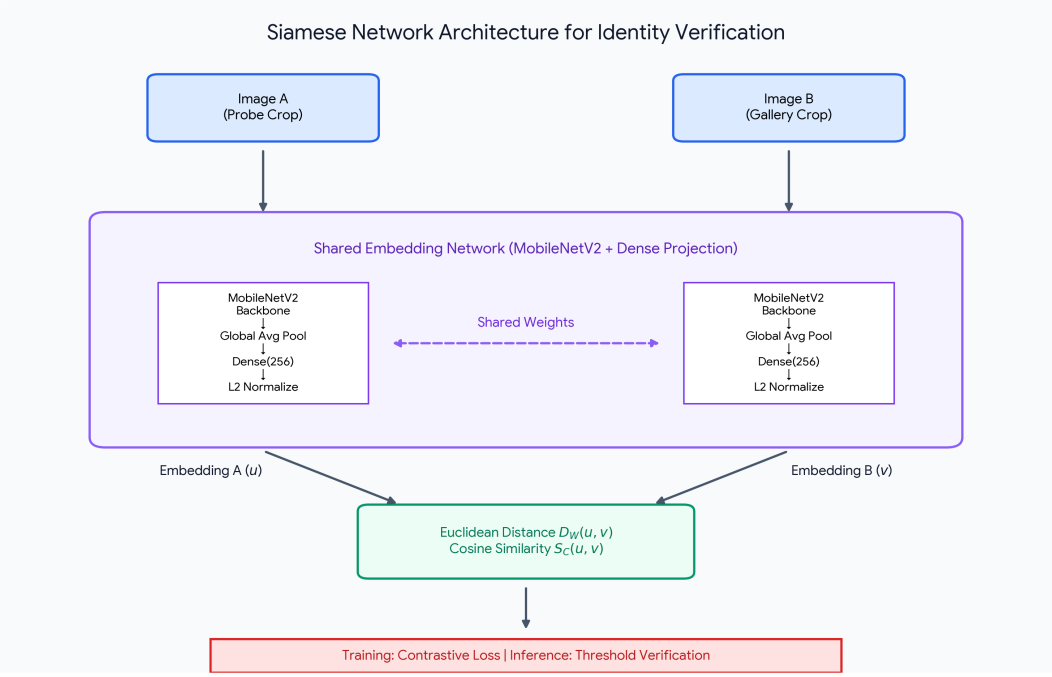


Figure 3: Siamese Network architecture using MobileNetV2 and contrastive loss

5 Mathematical Foundation

The proposed system combines convolutional neural network operations for face mask classification with a Siamese Network for face identity verification. This section presents the mathematical concepts underlying image preprocessing, feature extraction, embedding generation, activation functions, loss functions, and similarity measurement.

5.1 Image Preprocessing

Image preprocessing is performed before the input images are passed to the deep learning models. It ensures that the images have a consistent format, and numerical range. The preprocessing pipeline consists of noise and brightness estimation, and pixel normalization.

5.1.1 Noise and Brightness Assessment

To estimate the noise level, Gaussian smoothing is applied to the grayscale image using a 5×5 kernel:

$$I_s(x, y) = G_\sigma(x, y) * I_{\text{gray}}(x, y) \tag{1}$$

where G_σ is the Gaussian kernel and $*$ denotes convolution.

The residual image is obtained by subtracting the smoothed image from the original grayscale image:

$$R(x, y) = I_{\text{gray}}(x, y) - I_s(x, y) \tag{2}$$

The noise score is estimated using the Median Absolute Deviation (MAD):

$$\sigma_n = \frac{\text{median}(|R(x,y)|)}{0.6745} \tag{3}$$

The average brightness of the image is calculated as:

$$B = \frac{1}{WH} \sum_{x=1}^{W} \sum_{y=1}^{H} I_{\text{gray}}(x,y) \tag{4}$$

where W and H represent the width and height of the image.

Based on the thresholds used in the implementation, an image is rejected if its noise score exceeds 15, its average brightness is below 50, or its average brightness is above 220. Therefore, the image-quality decision is defined as:

$$Q(I) = \begin{cases} 1, & \sigma_n \leq 15 \text{ and } 50 \leq B \leq 220, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Here, $Q(I) = 1$ indicates that the image is accepted for further processing, whereas $Q(I) = 0$ indicates that it is rejected. This procedure estimates noise and filters images based on quality; it does not directly remove noise from the image.

5.1.2 Pixel Normalization

After resizing, pixel values are normalized to the range $[0, 1]$. For an image with 8-bit pixel values, each pixel intensity $p \in [0, 255]$ is transformed as:

$$p_{\text{norm}} = \frac{p}{255} \tag{6}$$

The normalized image is therefore represented as:

$$I_{\text{norm}} \in [0,1]^{224 \times 224 \times C} \tag{7}$$

Normalization reduces the scale of the input values and helps provide numerically stable inputs during model training and inference.

5.2 Convolution Operation

Convolution is a fundamental operation in CNNs used to extract local features from an image. A learnable filter slides over the input image and computes weighted sums of the values in each local region.

For a single-channel input image I, the convolution operation is:

$$Z(i,j) = \sum_{m=0}^{K_h-1} \sum_{n=0}^{K_w-1} I(i+m, j+n) K(m,n) + b \tag{8}$$

where:

- I is the input image or feature map,
- K is the convolution kernel,
- K_h and K_w are the kernel height and width,

- b is the bias,
- $Z(i, j)$ is the resulting value at position (i, j).

For multi-channel inputs, the operation also sums over the input channels:

$$Z(i, j, o) = \sum_{c=1}^{C_{\text{in}}} \sum_{m=0}^{K_h-1} \sum_{n=0}^{K_w-1} I(i+m, j+n, c) K(m, n, c, o) + b_o \tag{9}$$

Here, o denotes the output channel and C_{in} is the number of input channels.

5.3 Kernel or Filter Mechanism

A convolution kernel is a small matrix of learnable weights that detects patterns in a local image region. During training, the kernel weights are updated through backpropagation to learn useful visual features.

Different filters may respond to different patterns, such as edges, textures, contours, and more complex facial structures. As the network becomes deeper, the learned filters combine lower-level features to represent more abstract information.

In MobileNetV2, computational efficiency is improved through depthwise separable convolution, which separates spatial filtering from channel-wise feature combination.

5.3.1 Depthwise Convolution

Depthwise convolution applies one spatial filter to each input channel independently:

$$Z_{\text{dw}}(i, j, c) = \sum_{m} \sum_{n} I(i+m, j+n, c) K_{\text{dw}}(m, n, c) \tag{10}$$

This operation extracts spatial features while avoiding a full convolution across all input channels.

5.4 Activation Functions

Activation functions introduce non-linearity into the network, allowing it to learn complex relationships in the input data.

5.4.1 ReLU6 Activation

MobileNetV2 uses the ReLU6 activation in its feature extraction blocks. It is defined as:

$$f(x) = \min(\max(0, x), 6) \tag{11}$$

The function restricts the output to the range $[0, 6]$, providing non-linearity while limiting the activation values.

5.4.2 Sigmoid Activation

For binary face mask classification, a sigmoid activation can be used in the final output layer to produce a probability:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{12}$$

The output lies in the interval $(0, 1)$. Given a predicted probability p, the class is determined using a threshold τ:

$$\hat{y} = \begin{cases} 1, & p \geq \tau \\ 0, & p < \tau \end{cases} \tag{13}$$

For example, when $\tau = 0.5$, the image is assigned to one of the two classes according to the predicted probability.

5.5 Loss Functions

Loss functions measure the difference between the model's predictions and the expected outputs. The classification and identity-verification branches use different objectives.

5.5.1 Binary Cross-Entropy Loss

For binary face mask classification, binary cross-entropy is commonly used when the model produces a sigmoid probability. It is defined as:

$$L_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \tag{14}$$

where y_i is the ground-truth label and p_i is the predicted probability for sample i.

5.5.2 Contrastive Loss

The Siamese Network is trained using contrastive loss, which encourages embeddings of matching identity pairs to be close and embeddings of non-matching pairs to be separated by a margin.

The Euclidean distance between two embeddings is:

$$D = \|\mathbf{e}_1 - \mathbf{e}_2\|_2 \tag{15}$$

Using the convention that $Y = 0$ denotes a same-identity pair and $Y = 1$ denotes a different-identity pair, contrastive loss is:

$$L_{\text{contrastive}} = \frac{1}{2}(1 - Y)D^2 + \frac{1}{2}Y \max(0, m - D)^2 \tag{16}$$

where m is the margin parameter.

For matching pairs, the loss minimizes the embedding distance. For non-matching pairs, the loss penalizes distances smaller than the specified margin.

5.6 Cosine Similarity

Cosine similarity measures the angular similarity between two embedding vectors. It is defined as:

$$S_{\cos} = \frac{\mathbf{e}_1 \cdot \mathbf{e}_2}{\|\mathbf{e}_1\|_2 \|\mathbf{e}_2\|_2} \tag{17}$$

where $\mathbf{e}_1 \cdot \mathbf{e}_2$ is the dot product of the embeddings and $\|e_1\|_2$ and $\|e_2\|_2$ means the L2 norm, also called the Euclidean norm.

The cosine similarity value ranges from -1 to 1. A larger value indicates that the embeddings point in a more similar direction. Identity verification can be performed by comparing the similarity score with a threshold θ:

$$\text{Match} = \begin{cases} 1, & S_{\cos} \geq \theta \\ 0, & S_{\cos} < \theta \end{cases} \tag{18}$$

6 Input Specification

The proposed system accepts facial images as input for face mask classification and identity verification. The input images are processed through an image-quality assessment and preprocessing pipeline before being passed to the deep learning models.

6.1 Image Input

The system accepts digital facial images in common image formats such as JPEG and PNG. The images may contain individuals wearing a face mask or not wearing a face mask.

The input image is represented as:

$$I \in \mathbb{R}^{H \times W \times 3} \tag{19}$$

where H and W denote the original image height and width, and the three channels correspond to the color channels of the image.

6.2 Image Quality Requirements

Each input image is assessed for noise level and brightness. Images with excessive estimated noise, insufficient brightness, or overexposure are rejected based on the predefined quality thresholds.

The image is accepted when the following conditions are satisfied:

$$\sigma_n \leq 15, \qquad 50 \leq B \leq 220 \tag{20}$$

where σ_n is the estimated noise score and B is the average grayscale brightness.

6.3 Preprocessed Input for MobileNetV2

After quality assessment, the accepted image is resized to 224×224 pixels. The pixel values are normalized to the range $[0, 1]$.

The resulting input tensor is represented as:

$$I_{\text{CNN}} \in [0, 1]^{224 \times 224 \times 3} \tag{21}$$

This tensor is provided to the MobileNetV2-based CNN for binary face mask classification.

6.4 Dataset Used

The experiments in the project are conducted using the *A Masked and Unmasked Face Dataset*, available through Mendeley Data (Version 2). The dataset consists of facial images of individuals wearing face masks(Masked, M) and individuals without face masks (UnMasked, UM).
The dataset contains 660 images which includes images from 112 distinct human subjects. The class distribution is balanced as there are 330 masked images and 330 unmasked images.

The dataset is publicly available through Mendeley Data [1].

6.4.1 Dataset Split

The dataset is divided into training, validation, and testing subsets using a subject-wise splitting strategy. This ensures that images belonging to the same subject are assigned to only one subset, reducing identity overlap and helping provide a more reliable evaluation of the model. Table 1 shows the Dataset split and Pair distribution.

Table 1: Dataset split and verification pair distribution

Dataset Split	**Subject IDs**	**Total Images**	**Total Pairs**	**Pair Distribution**
Training	78	462	2,286	1,143 Pos / 1,143 Neg
Validation	17	100	492	246 Pos / 246 Neg
Testing	17	98	474	237 Pos / 237 Neg
Total	**112**	**660**	**3,252**	**1,626 Pos / 1,626 Neg**

6.4.2 Data Augmentation

Data augmentation is a technique used to increase the diversity of training samples by applying label-preserving transformations to existing images. It can help the model generalize to variations in facial appearance, orientation, and image conditions, while reducing the risk of overfitting to the training samples. The data augmentaion is only applied to the training data. In the training pipeline, augmentation includes operations such as small rotations, horizontal flipping, zooming, and brightness variation.

7 Implementation

The proposed system was implemented in Python 3.12.13 using TensorFlow and Keras for model development and training, OpenCV for image processing, NumPy for numerical

operations, Matplotlib for visualization, and Pandas and scikit-learn for data handling and evaluation. GPU acceleration was used for model training, with the dataset processed in batches to manage memory usage.

7.1 Training Configuration

The configuration parameters used for the three main components of the proposed system are summarized in Tables 2, 3, and 4.

Table 2: MediaPipe FaceLandmarker configuration

Parameter	**Value**
Model	MediaPipe FaceLandmarker
Model asset	`face_landmarker.task`
Running mode	IMAGE
Maximum faces	1
Detection confidence	0.5
Face presence confidence	0.5
Tracking confidence	0.5
Landmark regions	Eyes and nose
ROI extraction	Upper-face region
ROI output size	224×224 pixels

Table 3: MobileNetV2 mask classification configuration

Parameter	**Value**
Model	MobileNetV2
Pretrained weights	ImageNet
Input size	$224 \times 224 \times 3$
Backbone	Fully fine-tuned
Batch size	32
Epochs	15 (maximum)
Optimizer	Adam
Learning rate	1×10^{-5}
Loss function	Binary cross-entropy
Output activation	Sigmoid
Classification threshold	0.5
Dropout rate	0.35
Augmentation	Horizontal flip, rotation, zoom, contrast
Early stopping patience	5 epochs
LR reduction factor	0.5
LR reduction patience	2 epochs
Minimum learning rate	1×10^{-7}

Table 4: Siamese Network identity verification configuration

Parameter	**Value**
Feature extractor	MobileNetV2 (ImageNet pretrained)
Input pair	Two upper-face images
Input size	$224 \times 224 \times 3$ per image
Shared weights	Yes
Embedding dimension	256
Embedding normalization	L2 normalization
Distance metric	Euclidean distance
Loss function	Contrastive loss
Contrastive margin	1.0
Batch size	32
Epochs	50 (maximum)
Optimizer	Adam
Learning rate	1×10^{-5}
Hard-negative candidate pool	Top 10 similar identities
Training pairs	2,286
Validation pairs	492
Testing pairs	474
Pair distribution	Balanced positive and negative pairs
Augmentation	Flip, brightness, contrast, random crop, noise
Early stopping patience	5 epochs
LR reduction factor	0.5
LR reduction patience	2 epochs
Minimum learning rate	1×10^{-7}

7.2 Code Implementation

Due to the extensive length of the complete source code, it has not been included in this report. The full implementation of the proposed CNN-based face mask recognition system, including image preprocessing, model training, and identity verification, is available in a public GitHub repository [2]. The repository serves as a supplementary resource, providing access to the complete codebase for further reference and reproducibility.

8 Result

The proposed CNN-based face mask recognition system was evaluated using the test dataset, which was kept separate from the training and validation datasets. The system consists of two main components: a mask classification model and a Siamese neural network for face identity verification.

8.1 Mask Classifier Results

The mask classification model was evaluated on the test dataset containing 98 face images, consisting of 49 masked-face images and 49 unmasked-face images. The model was trained for 15 epochs using the MobileNetV2-based architecture. Its performance was evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

8.1.1 Training and Validation Accuracy

Figure 4 illustrates the training and validation accuracy over 15 epochs. The training accuracy increased from approximately 65% in the first epoch to 100% by the final epochs. The validation accuracy also improved rapidly, reaching approximately 99%. The close relationship between the two curves indicates that the model learned the classification patterns effectively and maintained strong performance on the validation dataset.

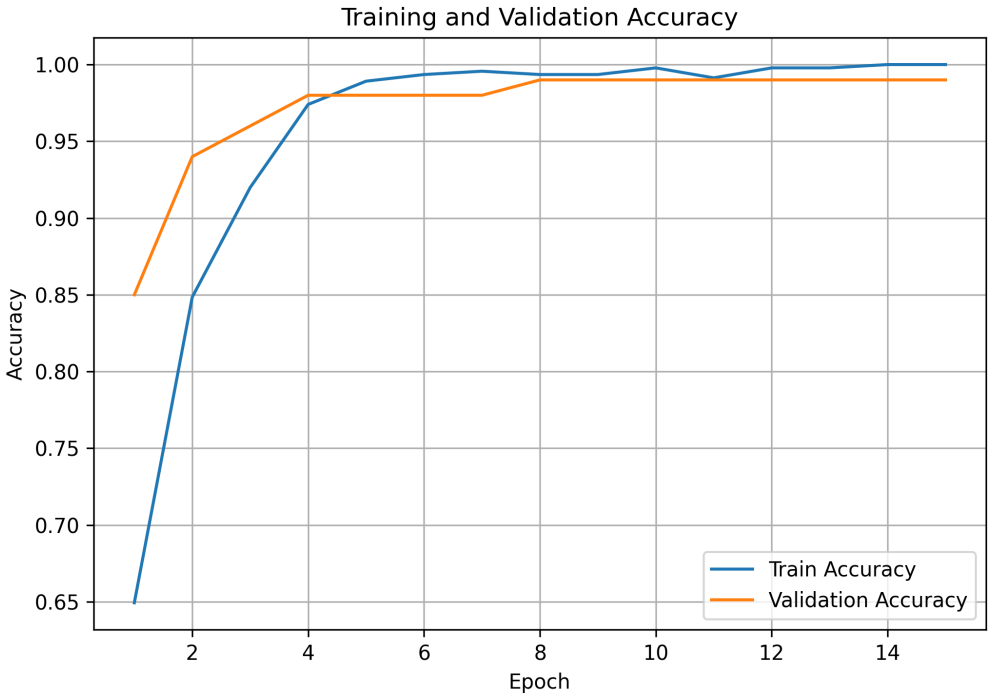


Figure 4: Training and validation accuracy of the mask classification model.

8.1.2 Training and Validation Loss

Figure 5 presents the training and validation loss over the training epochs. Both loss curves show a consistent downward trend. The training loss decreased from approximately 0.63 to 0.02, while the validation loss decreased from approximately 0.49 to 0.07. This reduction indicates that the model progressively minimized its classification error during training.

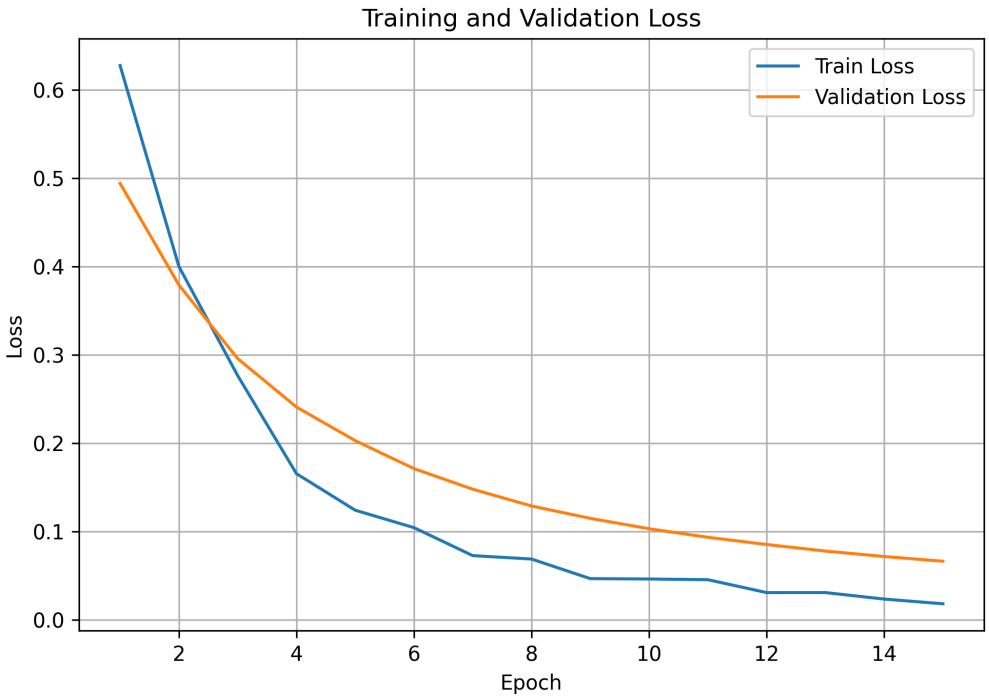


Figure 5: Training and validation loss of the mask classification model.

8.1.3 Confusion Matrix

The confusion matrix in Figure 6 shows the classification performance on the test dataset. Out of 49 masked images, all 49 were correctly classified as Masked. Similarly, all 49 unmasked images were correctly classified as UnMasked. There were no false positive or false negative predictions.

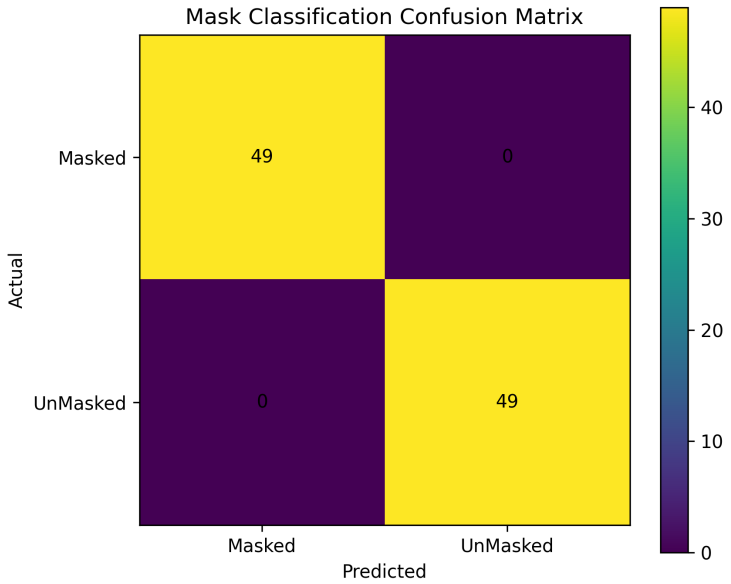


Figure 6: Confusion matrix for mask classification on the test dataset.

8.1.4 Classification Report and Evaluation Metrics

The classification report obtained from the test dataset is presented in Table 5. The model achieved 100% precision, recall, and F1-score for both classes.

Table 5: Mask classification performance on the test dataset.

Class	Precision	Recall	F1-score	Support
Masked	1.00	1.00	1.00	49
UnMasked	1.00	1.00	1.00	49
Accuracy	–	–	1.00	98
Macro Average	1.00	1.00	1.00	98
Weighted Average	1.00	1.00	1.00	98

The overall test accuracy, precision, recall, and F1-score are shown below:

```
========== TEST RESULTS ==========
accuracy  : 1.0000
precision : 1.0000
recall    : 1.0000
f1_score  : 1.0000
```

Thus, All 49 masked images and all 49 unmasked images were correctly classified. These results demonstrate excellent classification performance on the available test dataset.

8.2 Face Identity Verification Results

The face identity verification component was implemented using a Siamese neural network with a shared MobileNetV2-based feature extraction backbone. The network generates 256-dimensional feature embeddings for face images. These embeddings are compared using cosine similarity to determine whether two face images belong to the same individual or to different individuals.

The Siamese network was trained for 50 epochs using contrastive loss. Its performance was monitored using training and validation loss.

8.2.1 Training and Validation Loss

Figure 7 shows the training and validation contrastive loss over 50 epochs. The training loss decreased from approximately 0.33 to 0.07, while the validation loss decreased from approximately 0.35 to 0.17. Both curves show a gradual downward trend, indicating that the network learned to distinguish between positive pairs (same identity) and negative pairs (different identities).

The training loss remained lower than the validation loss throughout most of the training process. The difference between the two curves indicates a generalization gap between the training and validation datasets.

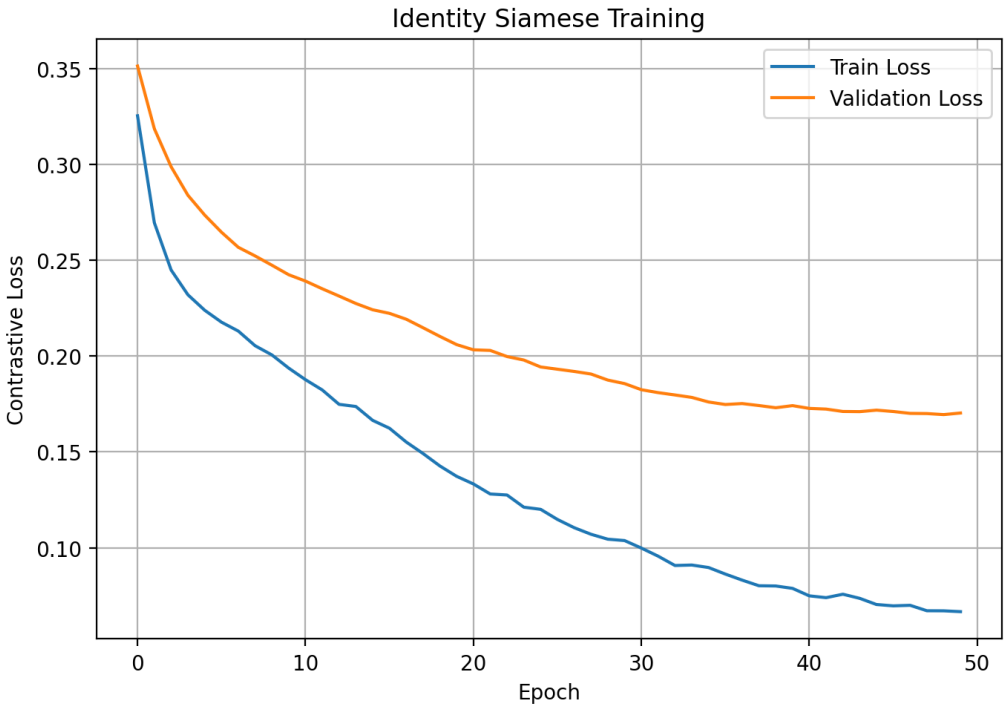


Figure 7: Training and validation contrastive loss of the Siamese network.

8.2.2 Validation-Based Threshold Selection

After training, cosine similarity was used to compare the feature embeddings of two face images. A similarity threshold was selected using the validation dataset by evaluating different threshold values and choosing the one that produced the best F1-score.

The identity verification output reported a validation-selected threshold of 0.8195 and a validation F1-score of 0.7984. A fixed threshold of 0.5 was also evaluated for comparison.

8.2.3 Test Dataset Evaluation

The Siamese network was evaluated using 474 test pairs, comprising 237 positive pairs (same identity) and 237 negative pairs (different identities). The test performance obtained using the validation-selected threshold is presented in Table 6.

Table 6: Identity verification performance on the test dataset.

Metric	**Validation Threshold**	**Fixed 0.5**
Threshold	0.8195	0.5000
Accuracy	76.37%	63.92%
Precision	74.51%	58.42%
Recall	80.17%	96.62%
F1-score	77.24%	72.81%
FAR	27.43%	68.78%

The validation-selected threshold achieved better accuracy, precision, and F1-score than the fixed threshold of 0.5. It also reduced the False Acceptance Rate (FAR) from 68.78% to 27.43%. Although the fixed threshold achieved higher recall, it incorrectly accepted a larger proportion of different-identity pairs.

8.2.4 Classification Report

The classification report evaluates the model's ability to distinguish between different-identity and same-identity pairs. The reported precision, recall, and F1-score for both classes are shown in Table 7.

Table 7: Classification report for identity verification.

Class	Precision	Recall	F1-score	Support
Different Identity	0.79	0.73	0.75	237
Same Identity	0.75	0.80	0.77	237
Accuracy	–	–	0.76	474
Macro Average	0.77	0.76	0.76	474
Weighted Average	0.77	0.76	0.76	474

8.2.5 Confusion Matrix

The confusion matrix provides a detailed representation of the correct and incorrect identity verification predictions. It shows the number of same-identity and different-identity pairs classified by the model.

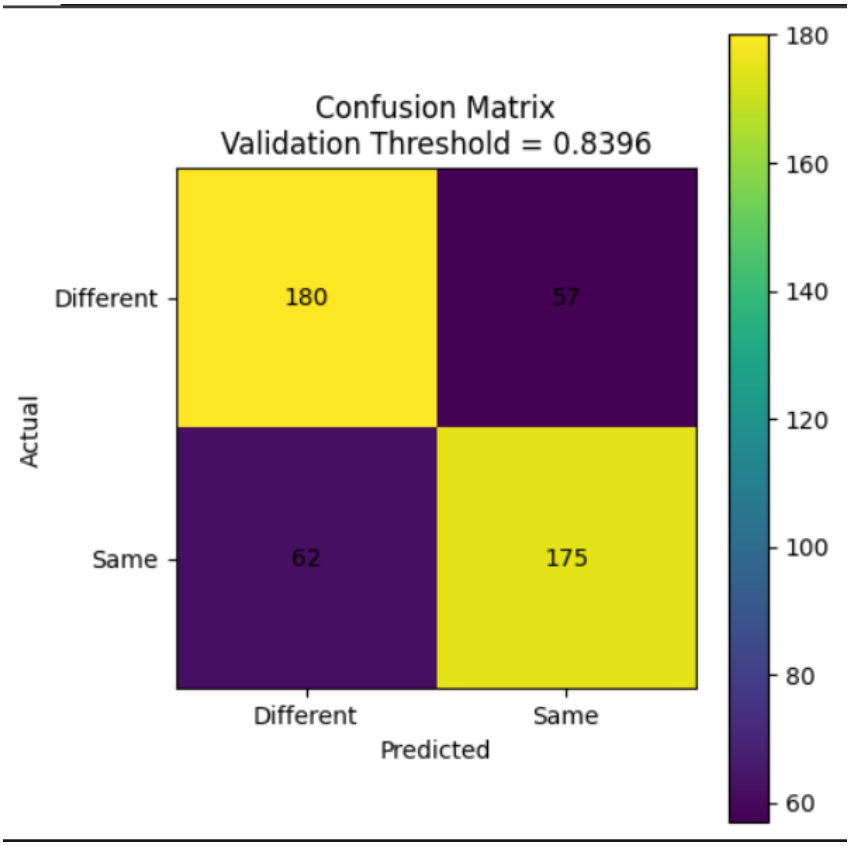


Figure 8: Confusion matrix for Siamese-based identity verification.

8.2.6 Overall Observation

The experimental results demonstrate that the Siamese network learned useful facial representations for identity verification. The decreasing training and validation losses

indicate progressive learning, while the test results show that selecting an appropriate similarity threshold improves verification performance.

8.3 Model Development and Performance Improvements

The proposed system was developed through several stages, with each version introducing improvements in model architecture, identity recognition, feature extraction, and evaluation. The major development stages are compared in Table 8.

Table 8: Comparison of major model development stages.

Model	Architecture	Loss Function	Improvement
Model A	Multi-task MobileNetV2	Categorical Cross-Entropy	Initial mask and identity classification.
Model F	MobileNetV2, 128-D embeddings	Triplet Loss	Introduced metric learning.
Model H	Siamese MobileNetV2, 128-D embeddings	Contrastive Loss	Improved face-pair verification.
Model I (Final)	Siamese MobileNetV2, 256-D embeddings, MediaPipe	BCE + Contrastive Loss	Added upper-face extraction and joint recognition.

The development process introduced several important improvements to make the system more reliable and suitable for identity verification under facial occlusion.

- **Replacement of Softmax Identity Classification with Metric Learning:** The initial model used a fixed Softmax identity classification layer. This approach was replaced with metric learning using Triplet Loss. The improvement allows the model to learn facial embeddings based on identity similarity rather than relying only on a fixed set of identity classes.

- **Introduction of Siamese Learning and Contrastive Loss:** Triplet Loss was later replaced with Contrastive Loss in the Siamese network. The network compares two face images and learns to minimize the distance between embeddings of the same person while increasing the distance between embeddings of different people. This provides a direct approach to face-pair verification.

- **Increase in Embedding Dimension:** The embedding representation was increased from 128 dimensions to 256 dimensions in the final model. This provides a larger feature representation for encoding visible facial characteristics.

- **MediaPipe-Based Upper-Face Extraction:** Earlier approaches relied on general face images. The final model introduced MediaPipe facial landmarks to extract the upper-face region, including the forehead, eyes, and nose. This helps the identity verification network focus on facial regions that remain visible when a person wears a mask.

- **Improved Identity Mapping and Dataset Splitting:** Identity mapping was corrected so that masked and unmasked images belonging to the same individual were assigned to the same identity. Identity-disjoint training, validation, and test

sets were also introduced to evaluate performance on previously unseen individuals and reduce data leakage.

- **Validation-Based Threshold Calibration:** Instead of relying only on a fixed similarity threshold, the final development introduced validation-based threshold selection. This helps balance false acceptance and false rejection during identity verification.

- **Separate Loss Functions for the Two Tasks:** The final system uses Binary Cross-Entropy for masked versus unmasked classification and Contrastive Loss for identity verification. This allows each component to optimize its respective learning objective.

Overall, the development progressed from conventional multi-task identity classification to embedding-based Siamese verification. The final model combines upper-face landmark extraction, a larger normalized embedding, and separate learning objectives for mask classification and identity verification. These changes were introduced to improve recognition under facial occlusion and make the evaluation more reliable.

9 Conclusion

9.1 Summary of Work

An end-to-end deep learning framework was developed for face mask classification and identity verification under facial occlusion. The proposed system combines a MobileNetV2-based CNN architecture with MediaPipe facial landmark detection and a Siamese neural network trained using Contrastive Loss.

The framework performs upper-face region extraction to focus on visible facial characteristics, followed by mask classification and identity verification. A 256-dimensional normalized feature embedding is generated for identity comparison. The system was trained and evaluated using an identity-disjoint dataset containing 112 subjects and 660 images.

9.2 Performance Analysis

- **Mask Classification:** The MobileNetV2-based classifier achieved 100% accuracy, 100% precision, and 100% recall on 98 test images, consisting of 49 masked and 49 unmasked images.

- **Identity Verification:** The Siamese network achieved 76.37% accuracy, 74.51% precision, 80.17% recall, and 77.24% F1-score on 474 test pairs. A validation-selected cosine similarity threshold of 0.8195 was used, resulting in a False Acceptance Rate (FAR) of 27.43%.

The results demonstrate strong performance in mask classification and promising identity verification capability under masked-face conditions. However, identity verification remains more challenging than mask classification and requires further improvement.

9.3 Limitations

1. **Limited Dataset Size:** The dataset contains 112 subjects and 660 images. A larger and more diverse dataset is required to evaluate the model's generalization across different individuals and environmental conditions.

2. **Facial Occlusion and Appearance Variations:** Masks, glasses, hairstyles, and other occlusions can reduce the amount of useful facial information available for identity verification.

3. **Pose and Landmark Detection:** Extreme head rotations and unusual facial poses may affect MediaPipe landmark detection and upper-face region extraction, potentially reducing verification accuracy.

4. **Identity Verification Errors:** The Siamese network still produces false acceptance and false rejection errors. Its verification performance requires further improvement before deployment in highly sensitive applications.

9.4 Future Improvements

1. **Improved Feature Extraction:** Investigate Vision Transformers and lightweight hybrid architectures to capture more discriminative facial features from partially visible faces.

2. **Larger and More Diverse Dataset:** Train and evaluate the system using additional identities, facial poses, lighting conditions, mask types, and occlusion variations.

3. **Improved Landmark-Based Alignment:** Investigate alternative facial anchor points and rotation-aware alignment methods to improve upper-face extraction under rotated and tilted facial poses. Using the nose as a primary anchor point may also be explored for handling faces rotated up to 90 degrees.

4. **Real-Time Webcam-Based Recognition:** Develop an integrated webcam-based application to capture live video frames and perform face detection, mask classification, and identity verification in real time. The system could continuously process detected faces and display their mask status and predicted identity, making it suitable for practical applications such as access control and attendance monitoring.

5. **Model Optimization and Deployment:** Apply TensorFlow Lite quantization and model optimization techniques to reduce computational requirements and support real-time deployment on edge devices.

10 Contributions

The specific contributions delivered in this project include:

1. **Identity-Disjoint Dataset Partitioning:** Implemented an identity-level dataset splitting strategy across 112 subjects, consisting of 78 training, 17 validation, and 17 testing subjects. This prevents identity overlap between the different data partitions.

2. **Automated Upper-Face Region Extraction:** Integrated MediaPipe Face Landmarker to detect facial landmarks and automatically extract the upper-face region. This allows the identity verification model to focus on visible facial information when the lower face is covered by a mask.

3. **Siamese-Based Identity Verification:** Developed a Siamese neural network using a shared MobileNetV2 backbone and Contrastive Loss. The network generates 256-dimensional normalized embeddings to compare facial images belonging to the same or different individuals.

4. **Hard Negative Mining Strategy:** Implemented hard negative mining using 32×32 grayscale image signatures and a top-K selection strategy with $K = 10$. This helps select visually similar images belonging to different identities during Siamese network training.

5. **Integrated Mask Classification and Identity Verification:** Combined a Binary Cross-Entropy-based mask classifier with a Contrastive Loss-based Siamese verification model into a unified framework.

6. **Validation-Based Threshold Calibration:** Implemented cosine similarity threshold selection using validation data and evaluated identity verification using accuracy, precision, recall, F1-score, and False Acceptance Rate.

11 References

References

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[2] Ghosh,A., & Biswas,R.(2026) *CNN-Based Face Mask Recognition System*, GitHub repository. Available at: https://github.com/thehiddenquest/CNN-Based-Face-Mask-Recognition-System.git